

HAYAYA, Comoros

WMO: 670020

Lat: 11.534S

Lon: 43.272E

Elev: 28

StdP: 100.99

Time Zone: 3.00 (E03)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.8	20.2	15.0	10.7	23.0	16.1	11.5	23.3	12.2	27.2	11.2	27.4	2.4	100	0.353

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.9	33.0	25.8	32.2	25.9	31.9	25.9	27.7	30.8	27.3	30.6	27.0	30.5	5.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.7	30.0	26.2	21.7	29.6	26.1	21.5	29.5	88.5	30.8	86.5	31.0	85.2	30.4	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.2	8.1	DB	18.4	34.4	0.4	0.5	18.1	34.8	17.8	35.1	17.6	35.3	17.3	35.7	(4)
(5)			WB	15.3	29.5	0.5	0.6	14.9	30.0	14.6	30.3	14.3	30.7	13.9	31.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.7	27.6	27.6	27.8	27.7	26.9	25.7	25.0	24.6	25.1	26.5	27.8	27.9	(6)	
	DBStd	1.58	1.00	0.95	0.88	0.96	1.01	0.96	0.98	0.99	1.09	1.34	1.21	0.92	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6094	547	493	551	532	524	472	465	453	454	512	534	556	(10)	
	CDD18.3	3052	288	259	292	282	266	222	207	195	204	254	284	298	(11)	
	CDH23.3	29183	3095	2750	3061	2912	2586	1822	1497	1373	1526	2288	3026	3248	(12)	
	CDH26.7	9303	1081	943	1091	1001	852	476	317	257	316	672	1089	1211	(13)	
(14) Wind	WSAvg	3.5	3.2	2.9	2.8	3.2	3.7	4.1	4.2	3.7	3.6	3.8	3.8	2.9	(14)	
(15) Precipitation	PrecAvg	1194	218	175	181	144	92	57	44	31	25	45	60	120	(15)	
	PrecMax	1888	427	344	373	269	242	94	79	63	106	170	282	354	(16)	
	PrecMin	775	78	63	84	59	31	29	25	18	4	16	14	22	(17)	
	PrecStd	299	85	72	78	60	47	19	13	10	21	30	52	79	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	32.7	33.1	33.0	32.2	31.2	30.4	30.5	31.0	32.9	34.0	33.2	(19)	
		MCWB	26.4	26.4	26.4	26.4	25.0	23.7	22.9	22.6	23.4	24.7	25.5	25.6	(20)	
	2%	DB	32.0	31.9	32.2	32.1	31.4	30.3	29.9	29.5	30.0	31.7	32.8	32.4	(21)	
		MCWB	26.6	26.5	26.5	26.1	25.0	23.8	22.8	22.8	23.3	24.6	25.3	25.9	(22)	
	5%	DB	31.2	31.2	31.6	31.5	30.9	29.8	29.1	28.9	29.1	30.8	31.9	31.8	(23)	
		MCWB	26.5	26.5	26.5	26.1	25.0	23.8	23.0	22.9	23.5	24.4	25.3	26.1	(24)	
	10%	DB	30.9	30.8	31.0	30.9	30.1	29.1	28.3	28.0	28.4	29.9	31.0	31.1	(25)	
		MCWB	26.4	26.4	26.5	26.1	24.8	23.8	23.0	22.9	23.4	24.4	25.5	26.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.3	27.9	28.1	27.8	27.0	26.5	26.0	25.5	25.7	27.1	27.9	27.8	(27)	
		MCDB	31.0	30.9	30.8	30.8	30.5	28.8	28.4	28.1	28.2	29.7	30.9	31.3	(28)	
	2%	WB	27.5	27.2	27.4	27.2	26.3	25.7	24.8	24.7	25.0	26.3	27.0	27.2	(29)	
		MCDB	30.7	30.5	30.6	30.7	30.1	28.7	27.9	27.7	28.0	29.3	30.4	30.9	(30)	
	5%	WB	26.9	26.9	27.0	26.8	25.8	25.0	24.2	24.1	24.5	25.8	26.5	26.7	(31)	
		MCDB	30.5	30.2	30.5	30.4	29.7	28.6	27.7	27.4	27.8	29.0	30.1	30.5	(32)	
	10%	WB	26.5	26.6	26.6	26.4	25.3	24.5	23.6	23.5	24.1	25.3	26.1	26.4	(33)	
		MCDB	29.9	29.8	30.1	29.9	29.3	28.1	27.3	27.1	27.5	28.7	29.9	30.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.3	6.8	6.8	7.5	7.6	7.7	7.9	7.5	7.2	7.0	6.9	(35)	
		MCDBR	7.1	7.1	7.5	7.6	8.0	8.4	8.4	8.6	8.1	7.9	7.8	7.6	(36)	
	5% WB	MCWBR	4.0	3.9	3.9	3.7	4.3	4.7	5.1	5.2	4.3	3.7	3.7	4.0	(37)	
		MCWBR	6.7	6.6	7.2	7.1	7.5	7.8	7.9	8.0	7.6	7.5	7.3	7.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.439	0.431	0.398	0.386	0.377	0.363	0.360	0.365	0.388	0.409	0.417	0.405	(40)	
		taud	2.449	2.475	2.590	2.613	2.608	2.618	2.613	2.585	2.497	2.462	2.461	2.537	(41)	
	Ebn at Noon		906	908	922	902	880	877	888	909	916	917	920	936	(42)	
		Edh at Noon	121	118	102	95	91	87	89	96	111	118	119	110	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.76	5.76	5.81	5.57	5.14	4.87	5.02	5.67	6.21	6.70	6.89	6.49	(44)	
	RadStd		0.57	0.47	0.35	0.36	0.24	0.18	0.14	0.12	0.18	0.19	0.27	0.39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page